NOTE

(i) $B(X: n, p)$ when $n \to \infty$ and p, q are not small will become a Normal distribution.

(i) $P(X, \lambda)$ when $\lambda \longrightarrow \infty$ will become a Normal distribution.

(ii) When $X \sim N(\mu, \sigma^2)$ then $Z = \frac{x-\mu}{\sigma}$ then $Z \sim N(0, 1)$

i.e., $f(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} \quad -\infty \le z \le \infty$

Z is known as a Standard Normal variate with mean 0 and variance 1.

To find probabilities of X we convert X into Z and then make use of standard normal table.

Properties of Normal Distribution

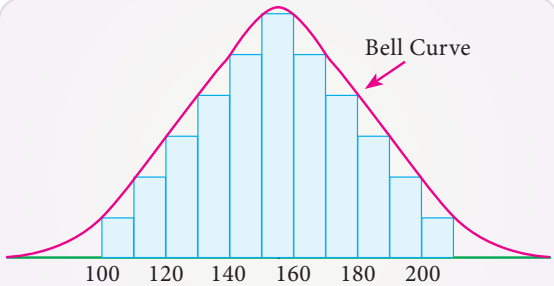
(i) The curve is bell shaped and is symmetric about $X = \mu$

(ii) Mean = Median = Mode = μ

(iii) Unimodal at $X = \mu$ and Model height $= \frac{1}{\sigma\sqrt{2\pi}}$

(iv) Skewness $\beta_1 = 0$ and Kurtosis $\beta_2 = 3$

(v) The points of inflections are at $x = \mu \pm \sigma$



Bell Curve
Fig.10.1

(vi) The X – axis is the asymptote to the curve

(vii) The mean deviation about mean is 0.8σ

(viii) Quartile deviation is 0.6745σ

(ix) If $X \sim N(\mu_1, \sigma_1^2,)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and X and Y are independent, then $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$


Fig.10.2

(x) Area properties:

(a) Total area $P(-\infty < X < \infty) = 1$

(b) Area about $X = \mu$: $P(-\infty < X < \infty) = P(\mu < X < \infty) = 0.5$